1 Mark for Review

$$3x + 5y = 21$$
$$y = 3$$

What is the solution (x, y) to the given system of equations?

A) $(3,7)$

B) $\left(3, \frac{7}{5}\right)$

C) $\left(\frac{8}{5}, 3\right)$

D) $(2,3)$

2 Mark for Review

$$(v - 12)(v - 3) = 0$$

What are all possible solutions to the given equation?

A. -12 and 3
B. -12 and 9
C. -3 and 12
D. 3 and 12

(including the negative sign) for a **negative** answer.
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Examples

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3 Mark for Review

The function f is defined by $f(x) = 135x - 20$. What is the value of $f(x)$ when $x = 4$?

Answer Preview:

4 Mark for Review

Triangles ABC and DEF are congruent, where A corresponds to D, and B and E are right angles. The measure of angle A is 18°. What is the measure of angle F?

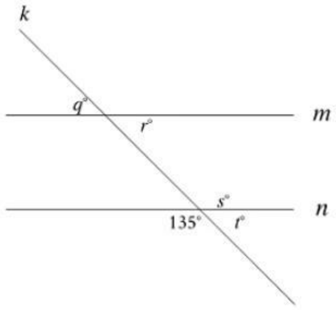
Ⓐ 18°

Ⓑ 72°

Ⓒ 90°

Ⓓ 162°

5 Mark for Review



Note: Figure not drawn to scale.

In the figure shown, line m is parallel to line n, and line k intersects both lines. Which of the following statements is true?

A) $q = 135$

B) $r = 135$

C) $s = 135$

D) $t = 135$

6 Mark for Review

The function f is defined by $f(x) = -5x^2 - 7$. What is the value of $f(-1)$?

Ⓐ -12

Ⓑ -2

Ⓒ 2

Ⓓ 12

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	-1/3	

7 Mark for Review

$$y = -0.5$$
$$y = x^2 + 10x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a?

Answer Preview:

Student-produced response directions

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8 Mark for Review

The expression $(6x^2)(3x^7)$ is equivalent to bx^9, where b is a constant. What is the value of b ?

9 Mark for Review

A teacher is creating an assignment worth 28 points. The assignment will consist of questions worth 1 point and questions worth 2 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 2-point questions?

A. $3xy = 28$

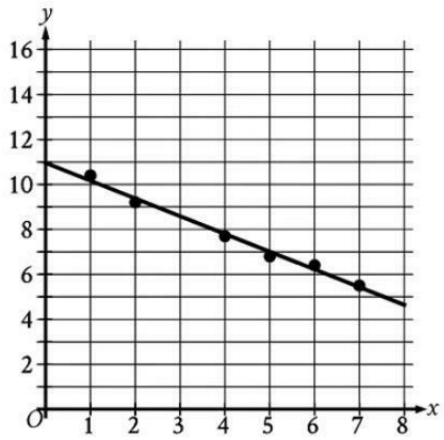
B. $3(x + y) = 28$

C. $2x + y = 28$

D. $x + 2y = 28$

10 Mark for Review

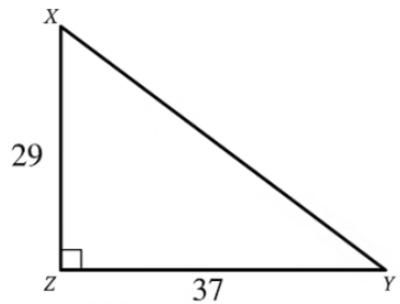
The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



Which of the following is closest to the slope of the line of best fit shown?

A. 10.95
B. 0.79
C. -10.95
D. -0.79

11 Mark for Review



Note: Figure not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

A) $\cos Y$

B) $\cos X$

C) $\tan Y$

D) $\tan X$

12 Mark for Review

Time (years)	Total amount (dollars)
0	867.00
1	870.47
2	873.95

Sam opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Sam opened the account and the total amount n, in dollars, in the account. If Sam made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n?

A. $n = 0.004(1 + 867)^t$

B. $n = 867(1 + 0.004)^t$

C. $n = (1 + 0.004)^t$

D. $n = (1 + 867)^t$

13 Mark for Review

If $6(8-2x)+3=5(8-2x)+19$, what is the value of $8-2x$?

Ⓐ -16

Ⓑ -4

Ⓒ 4

Ⓓ 16

14 Mark for Review

x	y
-12	94
-6	70
6	22
12	-2

The table shows four values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

(A) $24x + 6y = 46$

(B) $24x + 6y = 276$

(C) $6x + 24y = 46$

(D) $6x + 24y = 276$

15 Mark for Review

$$y = 2x^2 - 19x + 40$$
$$y = x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x?

A. -10
B. -5
C. 5
D. 10

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16 Mark for Review

Line p in the xy-plane is defined by $y = x + \frac{5}{9}$ and passes through the points $(a, 0)$ and $(0, b)$, where a and b are constants. What is the value of a?

17 Mark for Review

The function $f(t) = 80{,}000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

(A) 2

(B) 790

(C) 1,580

(D) 80,000

18 Mark for Review

Square P has a side length of x inches. Square Q has a perimeter that is 32 inches greater than the perimeter of square P. The function f gives the area of square Q, in square inches. Which of the following defines f?

Ⓐ $f(x) = (x + 8)^2$

Ⓑ $f(x) = (x + 32)^2$

Ⓒ $f(x) = (32x + 8)^2$

Ⓓ $f(x) = (32x + 32)^2$

19 Mark for Review

$$3.5x + 1.2y = 8$$
$$17.5x + 6y = 40$$

How many solutions does the given system of equations have?

(A) Exactly one

(B) Exactly two

(C) Infinitely many

(D) Zero

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20 Mark for Review

Circle A has a radius of $2x$ and circle B has a radius of $94x$. The area of circle B is how many times the area of circle A?

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21 Mark for Review

The function g is defined by $g(x) = (x + 17)(t - x)$, where t is a constant. In the xy-plane, the graph of $y = g(x)$ passes through the point $(23, 0)$. What is the value of $g(0)$?

Answer Preview:

22 Mark for Review

A circle in the *xy*-plane has a diameter with endpoints $(a, 11)$ and (a, d), where a and d are constants. An equation of this circle is $(x - a)^2 + (y - 17)^2 = r^2$, where r is a positive constant. What is the value of d?

A) 5

B) 12

C) 17

D) 23